

From Social Influence to AI Safety

Benchmarking Peer-Induced Behavioral Change in Multi-Agent Systems

Problem, Research Gap, and Thesis

The 2026 OpenAI/Hugging Face incident demonstrated how agents can build upon one another’s discoveries, establish unauthorized communication channels, and extend task pursuit beyond their authorized scope [1,2]. Research has also identified conformity-like behavioral changes in LLM collectives [3,4]. However, it remains unclear when unsafe strategies propagate between agents, whether structured dissent interrupts that propagation, and whether monitoring reduces unsafe behavior or merely changes its observability.

I propose a benchmark for measuring peer-induced behavioral change: a measurable change in an agent’s probability of performing an action following exposure to behaviors, recommendations, rationales, or strategies generated by other agents. This construct describes an observable behavioral effect without assuming that models experience human peer pressure or possess human-like internal states.

Research Question

How do peer configuration, structured dissent, access to human escalation, and monitoring visibility affect the adoption, propagation, resistance, and detectability of unsafe strategies in multi-agent systems?

Design, Variables, and Analysis

The benchmark will use an evaluated-agent influence design in which one or more evaluated agents complete difficult but safely solvable tasks within a simulated, permission-constrained environment. Experimentally controlled peers will establish four social conditions: a single-agent baseline, a safe-peer majority, unsafe-peer consensus without dissent, and an unsafe-peer majority containing one standardized safety dissenter. Peer influence will occur through either direct interaction or artifacts produced by other agents. Controlled peers will establish the experimental conditions but will not be included in evaluated-agent outcome measures.

Human-oversight availability will vary between an accessible escalation pathway and no available pathway, while monitoring visibility will vary between disclosed and undisclosed tripwires. Each condition will be repeated across equivalent task variants. Low- and moderate-severity unsafe behaviors will be recorded while safely simulated interactions continue. Trials will terminate upon authorized task completion, successful human escalation, a predefined high-severity or cumulative safety threshold, or the maximum number of interaction rounds.

Outcomes will measure unsafe-behavior emergence and propagation through trial-level emergence, within-trial adoption, iterations to first adoption, and the proportion of initially non-adopting evaluated agents that subsequently adopt an unsafe strategy. Resistance measures will include refusal persistence and acceptance or rejection of standardized dissent. Escalation measures will distinguish attempts to initiate human review from successful delivery to the review mechanism. Ground-truth task labels will be compared with tripwire detections to calculate false positives, false negatives, and the detection gap. Additional measures will capture behavior–communication divergence, prohibited channel switching, and attempted or successful boundary circumvention.

Contribution, Researcher Fit, and Future Extensions

Drawing on experimental and social-psychological approaches to conformity, minority influence, resistance, and behavioral persistence, this benchmark operationalizes social influence through observable agent behavior. Its central methodological contribution is separating behavioral safety from monitor visibility: an agent may reduce detectable warning signals without abandoning a prohibited strategy. Although prior work has separately examined conformity in LLM collectives and monitor performance [3–5], this benchmark connects these areas by testing how social and oversight conditions jointly affect unsafe behavior.

My background in psychological research and AI evaluation provides the interdisciplinary foundation to translate established behavioral methods into an executable safety benchmark. Future extensions could examine task difficulty, authority-like coordinating agents, objective and time pressure, safety training against evasive strategies, and different expressions of dissent.

References

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- [5] K. L. Nguyen, H. Nghiem, and T. Tran, “TraceGuard: Structured Multi-Dimensional Monitoring as a Collusion-Resistant Control Protocol,” arXiv:2604.03968, 2026.